Markov Chain Monte Carlo Algorithms

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1 Introduction

MCMC is a collection of methods that produce a realization of a Markov chain in order to estimate features of a given target distribution F. That is, the goals are the same as in the previous chapter, but the sample is no longer iid. The focus here is on introducing the foundational MCMC algorithms and a few of the properties of Markov chains that are required for them to eventually produce a representative sampler from F.

2 Markov Chains

Let $(\mathsf{X}, \mathcal{B})$ be a measurable space. A sequence of X-valued random variables $\{X_1, X_2, X_3, \ldots\}$ is a Markov chain if for all g

$$E\left[g(X_{n+1}, X_{n+2}, \ldots) \mid X_n, X_{n-1}, \ldots, X_1\right] = E\left[g(X_{n+1}, X_{n+2}, \ldots) \mid X_n\right].$$

Then P is a *Markov kernel* if $P : \mathsf{X} \times \mathcal{B} \to \mathbb{R}$ satisfying (i) for each fixed $x \in \mathsf{X}$, $P(x, \cdot)$ is a probability measure and (ii) for each fixed $B \in \mathcal{B}$, $P(\cdot, B)$ is a measurable function.

When X is a discrete set a Markov kernel can be represented as a square matrix whose entries are nonnegative and whose rows sum to 1.

Example 2.1. Suppose

$$P = \begin{pmatrix} 1/2 & 1/2 \\ 1/3 & 2/3 \end{pmatrix}$$

Then P is a Markov matrix on two states $\{0, 1\}$, say. The first row, for example, is interpreted as the probability of moving in one step from state 0 to state 0 is 1/2 which is the same as the probability of moving in one step from state 0 to state 1.

Example 2.2. Let $\mathsf{X} = \mathbb{Z}$ and let $0 < \theta < 1$. If $x \geq 1$, then a Markov kernel is defined by the matrix P with elements

$$P(x, x+1) = P(-x, -x-1) = \theta, \qquad P(x, 0) = P(-x, 0) = 1 - \theta,$$

and $P(0, 1) = P(1, 0) = 1/2$.

Often X will be uncountable and $\mathcal{B}$ will be countably generated. If X is topological, then $\mathcal{B}$ will be the Borel σ-algebra generated by X.

Example 2.3. Let $\mathsf{X} = (0,1)$ and consider the Markov chain that evolves as follows. Draw $U \sim$ Uniform$(0,1)$. If $u \le 0.5$, $X_{n+1} \sim$ Uniform$(0, X_n)$, but if $u > 0.5$, $X_{n+1} \sim$ Uniform$(X_n, 1)$. If $X_n = x$ and $B \in \mathcal{B}$, then

$$P(x,B) = \int_B \left[\frac{1}{2}\frac{1}{x} I_y((0,x)) + \frac{1}{2}\frac{1}{1-x} I_y(x,1) \right] \mathrm{d}y.$$

In Example 2.3 the integrand in the Markov kernel is a conditional density on X. If there is a conditional pf, say $k(y \mid x)$, with respect to a measure λ, such that the Markov kernel satisfies for $B \in \mathcal{B}$

$$P(x,B) = \int_B k(y \mid x) \lambda(\mathrm{d}y),$$

then k is a *Markov transition density.*

Example 2.4. Suppose $f(x,y)$ has support $\mathbb{R}^2$ with conditionals $f_{X|Y}(x \mid y)$ and $f_{Y|X}(y \mid x)$. Then, it is easy to show that

$$k(x', y' \mid x, y) = f_{X|Y}(x' \mid y) f_{Y|X}(y' \mid x')$$

is a Markov transition density. The Markov chain evolves from $(X_k = x, Y_k = y)$ to (X_{k+1}, Y_{k+1}) by drawing $X_{k+1} \sim F_{X|Y}(\cdot \mid y)$ followed by $Y_{k+1} \sim F_{Y|X}(\cdot \mid X_{k+1})$. This is a special case of the two-variable Gibbs sampler.

Suppose λ is a positive measure on $(\mathsf{X}, \mathcal{B})$, define

$$\lambda P(B) = \int_{\mathsf{X}} \lambda(\mathrm{d}x) P(x,B). \tag{1}$$

When λ is a probability measure, the encouraged interpretation is that $X_{n+1} \mid X_n \sim P(X_n, \cdot)$ and $X_n \sim \lambda$, the product $\lambda(\mathrm{d}x)P(x,\cdot)$ is the joint distribution of (X_n, X_{n+1}) and λP is the marginal distribution of X_{n+1}.

Since Markov kernels act to the left on measures as in equation (1),

$$P^2(x,B) = \int_{\mathsf{X}} P(x, \mathrm{d}x_k) P(x_k, B).$$

Continuing in this fashion obtain for every $n \geq 2$

$$P^n(x, B) \int_{\mathsf{X}} P(x, \mathrm{d}x_k) P(x_k, \mathrm{d}x_{k+1}) \cdots P(x_{k+n-2}, B).$$

More generally, the so-called Chapman-Kolmogorov equations hold for $n \geq m \geq 0$

$$P^n(x, B) \int_{\mathsf{X}} P^m(x, \mathrm{d}y) P^{n-m}(y, B).$$

If $\lambda = \lambda P$, then λ is *invariant* for P. Notice that if λ is invariant for P and $X_n \sim \lambda$, then $X_{n+1} \sim \lambda$. That is, the marginal distribution does not depend upon n in which case the Markov chain is *stationary*.

Example 2.5. Recall the Gibbs sampler of Example 2.4 having Markov transition density

$$k(x', y' \mid x, y) = f_{X|Y}(x' \mid y) f_{Y|X}(y' \mid x').$$

Routine calculation yields

$$\begin{aligned}
\iint f(x, y) k(x', y' \mid x, y) \mathrm{d}x dy &= \iint f(x, y) f_{X|Y}(x' \mid y) f_{Y|X}(y' \mid x') \mathrm{d}x \mathrm{d}y \\
&= f_{Y|X}(y' \mid x') \int f_{X|Y}(x' \mid y) \int f(x, y) \mathrm{d}x \mathrm{d}y \\
&= f_{Y|X}(y' \mid x') \int f_{X|Y}(x' \mid y) f_Y(y) \mathrm{d}y \\
&= f_{Y|X}(y' \mid x') f_X(x') \\
&= f(x', y').
\end{aligned}$$

Hence the joint density $f(x, y)$ is invariant for the Markov kernel.

One common way of establishing invariance of MCMC Markov chains is to verify a *detailed balance condition*

$$\lambda(\mathrm{d}x) P(x, \mathrm{d}y) = \lambda(\mathrm{d}y) P(y, \mathrm{d}x). \tag{2}$$

Indeed, Metropolis-Hastings Markov chains satisfy detailed balance by construction. To see that detailed balance implies λ is invariant for P, integrate both sides of the equation. The details are left as an exercise.

When λ is a probability measure, one interpretation is that the joint distribution of (X_k, X_{k+1}) is the same as the distribution of (X_{k+1}, X_k) so that this is also often called the *reversibility condition*. Another name often encountered is that P is λ-*symmetric*.

2.1 Stability

MCMC applications are constructed so that a specific probability distribution F is invariant. However, in applications where MCMC is required it is typically difficult to simulate from the invariant distribution. The most that can be hoped for is that the simulation will eventually produce a representative sample from F. This long-run behavior is in not guaranteed without additional assumptions. The following simple examples illustrate that the problems can arise due to the either the way the kernel is specified or the properties of the state space X.

Example 2.6. Suppose F lives on $\{1,2\}$ with $F(1) = 1 - F(2) = 1/4$ and

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Since the Markov chain moves deterministically between the two states, it will overrepresent state 1 and underrepresent state 2 no matter how many iterations there are.

Example 2.7. Suppose F lives on $\{1,2,3\}$ with $F(1) = F(2) = F(3) = 1/3$ and

$$P = \begin{pmatrix} 1/2 & 1/2 & 0 \\ 1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Thus starting at state $\{3\}$ the chain remains there forever while starting from $\{1,2\}$ the chain will never visit $\{3\}$. Thus the chain cannot represent F in the long run even though F is invariant for P.

Example 2.8. Suppose for $i = 1, 2$, f_i is a pf on $\mathsf{X}_i \subseteq \mathbb{R}$ and g_i is a pf on $\mathsf{Y}_i \subseteq \mathbb{R}$. Set

$$f(x,y) = \frac{1}{2} f_1(x) g_1(y) + \frac{1}{2} f_2(x) g_2(y).$$

Then

$$f_{X|Y}(x \mid y) = \frac{f_1(x)g_1(y) + f_2(x)g_2(y)}{g_1(y) + g_2(y)}$$

and

$$f_{Y|X}(y \mid x) = \frac{f_1(x)g_1(y) + f_2(x)g_2(y)}{f_1(y) + f_2(y)}$$

and the Gibbs sampler MTD is

$$k(x', y' \mid x, y) = f_{X|Y}(x' \mid y) f_{Y|X}(y' \mid x').$$

When $\mathsf{X}_i = \mathsf{Y}_i = \mathbb{R}$ this Gibbs sampler will produce a representative sample eventually. However, complications may arise if the spaces are constrained.

Suppose $\mathsf{X}_1 = \mathsf{X}_2 = (0, 1)$ and that $f_1 = f_2$ is the Uniform density. Let $Y_1 = (0, 1)$ and $Y_2 = (2, 3)$ and g_1 and g_2 be Uniform densities. Easy calculation yields that

$$f_{X|Y}(x \mid y) = I(0 < x < 1) \left[I(0 < y < 1) + I(2 < y < 3) \right]$$

and

$$f_{Y|X}(y \mid x) = \frac{1}{2} I(0 < x < 1) \left[I(0 < y < 1) + I(2 < y < 3) \right]$$

and that, no matter which square, $\mathsf{X}_1 \times \mathsf{Y}_1$ or $\mathsf{X}_2 \times \mathsf{Y}_2$, the current state is in there is a positive probability of the next state being in either square. This Gibbs sampler will eventually produce a representative sample.

Now consider the setting with $\mathsf{X}_1 = \mathsf{Y}_1 = (0, 1)$ and $\mathsf{X}_2 = \mathsf{Y}_2 = (2, 3)$ so that

$$f_{X|Y}(x \mid y) = \frac{I(0 < x < 1) I(0 < y < 1) + I(2 < x < 3) I(2 < y < 3)}{I(0 < y < 1) + I(2 < y < 3)}$$

and

$$f_{Y|X}(y \mid x) = \frac{I(0 < x < 1) I(0 < y < 1) + I(2 < x < 3) I(2 < y < 3)}{I(0 < x < 1) + I(2 < x < 3)}.$$

If $y \in (0, 1)$, then $f_{X|Y}(x \mid y) = I(0 < x < 1)$ and, similarly, if $X \in (0, 1)$, then $f_{Y|X}(y \mid x) = I(0 < y < 1)$. Thus if the current state is in the square $\mathsf{X}_1 \times \mathsf{Y}_1$, then the next step will be in $\mathsf{X}_1 \times \mathsf{Y}_1$. That is, there is no chance for the chain to visit $\mathsf{X}_2 \times \mathsf{Y}_2$. This Gibbs sampler will not produce a representative sample from the target distribution.

The above examples demonstrate that one way problems arise is when the Markov chain cannot access all of the space eventually and hence properties which avoid this are required.

Let ϕ be a non-trivial positive measure on $\mathcal{B}$. Then $A \in \mathcal{B}$ is *ϕ-communicating* if for all $B \subseteq A$ such that $\phi(B) > 0$ and for all $x \in A$, there exists n such that $P^n(x, B) > 0$. This is a weak property that does not alone ensure desirable long run properties. Consider the Gibbs sampler from Example 2.8 with $\mathsf{X}_1 = \mathsf{Y}_1 = (0,1)$ and $\mathsf{X}_2 = \mathsf{Y}_2 = (2,3)$. If ϕ denotes Lebesgue measure, then this Markov chain is ϕ-communicating, but does not have desirable long run properties.

The Markov kernel, P, is *ϕ-irreducible* if for all $x \in \mathsf{X}$ and for all $A \in \mathcal{B}$ such that $\phi(A) > 0$ there exists n such that $P^n(x,A) > 0$. This is a key property in many MCMC settings, but is not enough to ensure desirable long run properties; consider the Markov chain in Example 2.6 which is irreducible.

There is some arbitrariness in the definition of ϕ-irreducibility, but if for some ϕ, P is ϕ-irreducible, then there exists a maximal irreducibility measure ψ (meaning that $\psi(A) = 0$ implies $\phi(A) = 0$ for all irreducibility measures ϕ).

Proposition 2.1. *If λ is an invariant measure for the Markov kernel P and, for some ϕ, P is ϕ-irreducible, then P is λ-irreducible.*

Corollary 2.1. *If P is λ-symmetric and ϕ-irreducible, then P is λ-irreducible.*

Often, inspection of the kernel P and the underlying space X is enough to establish ϕ-irreducibility. For example, many kernels obviously satisfy a positivity condition so that $P(x,A) > 0$ for all $x \in \mathsf{X}$ and $A \in \mathcal{B}$ with $\phi(A) > 0$. The following is a special case of a more general result [4, Theorem 3].

Proposition 2.2. *Suppose X is a connected, separable metric space. If every nonempty, open set A satisfies $\phi(A) > 0$ and every point has a ϕ-communicating neighborhood, the Markov kernel is ϕ-irreducible.*

3 Metropolis-Hastings

The fundamental MCMC algorithm is the Metropolis-Hastings algorithm [6, 10]. It serves as a building block for many, if not most, applications of MCMC. The following will be generalized later, but, for now, suppose $\mathsf{X} \subseteq \mathbb{R}^d$ and that $q(y \mid x)$ is a proposal conditional density that is easy to sample. Define the *Hastings ratio*

$$r(x, y) = \frac{f(y)q(x \mid y)}{f(x)q(y \mid x)}.$$

Algorithm 1 Metropolis-Hastings

1: *Input:* Current value $X_n = x$.

2: Draw $Y \sim Q(x, \cdot)$

3: Draw $U \sim \text{Uniform}(0, 1)$

4: If $u \leq r(x, y) \wedge 1$, accept y and set $X_{n+1} = y$, else set $X_{n+1} = x$.

Algorithm 2 is the formal definition, but is not at how the algorithm should be implemented. The following implementation will help avoid overflow issues.

Algorithm 2 Metropolis-Hastings Implementation

1: *Input:* Current value $X_n = x$.

2: Draw $Y \sim Q(x, \cdot)$

3: Draw $U \sim \text{Uniform}(0, 1)$

4: If $\log r(x, y) \geq 0$ set $X_{n+1} = y$, else if $u \leq r(x, y)$, set $X_{n+1} = y$, else set $X_{n+1} = x$.

A Metropolis-Hastings algorithm defines a Markov kernel. Specifically, if $\alpha(x, y) = 1 \wedge r(x, y)$, then

$$P(x, \mathrm{d}y) = Q(x, \mathrm{d}y) + \delta_x(\mathrm{d}y) \int [1 - \alpha(x, u)]\, Q(x, \mathrm{d}u). \tag{3}$$

It is left as an exercise to establish this claim: Note that if $A \in \mathcal{B}$ and $j \geq 1$ then

$$\Pr(X_{j+1} \in A \mid X_j) = \Pr(X_{j+1} \in A, U \leq \alpha(x, y) \mid X_j) + \Pr(X_{j+1} \in A, U > \alpha(x, y) \mid X_j).$$

Proposition 3.1. *The Metropolis-Hastings kernel* (3) *is F-symmetric.*

Proof. It suffices to consider $x \neq y$. Then

$$\begin{aligned} F(\mathrm{d}x)P(x,\mathrm{d}y) &= f(x)q(y \mid x)\,[1 \wedge r(x,y)]\,\mathrm{d}x\mathrm{d}y \\ &= [f(x)q(y \mid x) \wedge f(y)q(x \mid y)]\,\mathrm{d}x\mathrm{d}y \\ &= f(y)q(x \mid y)\,[1 \wedge r(y,x)]\,\mathrm{d}x\mathrm{d}y \\ &= F(\mathrm{d}y)P(y,\mathrm{d}x). \end{aligned}$$

$\square$

The proof of the following result is left as an exercise.

Proposition 3.2. *If $q(y \mid x) > 0$ for all $x, y \in \mathsf{X}$, then the Metropolis-Hastings kernel P is F-irreducible.*

Example 3.1. Suppose continuous density f has support $\mathsf{X} = \mathbb{R}$ and consider three settings.

1. If the proposal distribution is the Student's t distribution with $d \geq 1$ degrees of freedom, then by Proposition 3.2 the resulting Metropolis-Hastings kernel is F-irreducible.

2. If the proposal distribution is Uniform$(0, 1)$, then the resulting Metropolis-Hastings kernel is obviously reducible since no value outside of $(0, 1)$ will be proposed.

3. If $X_n = x$ and the proposal is Uniform$(x - 1, x + 1)$, then Proposition 2.2 can be used to establish F-irreducibility. The first two conditions of the proposition are easy consequences of the properties of the real line. Let $0 < \delta < 1$ so that $N_{x,\delta} = (x - \delta, x + \delta)$ is a neighborhood of $\{x\}$ and
$$P(x, N_{x,\delta}) = \int_{x-\delta}^{x+\delta} \frac{1}{2}\left(1 \wedge e^{-0.5(x^2+y^2)}\right) \mathrm{d}y > 0$$
from which it is easy to see that $N_{x,\delta}$ is an F-communicating neighborhood.

3.1 Classical Strategies for Choosing a Proposal

An *Independence Metropolis-Hastings* algorithm results if the proposal is independent of the previous state. That is, $Y \sim Q$. For example, suppose the proposal is a d-dimensional normal distribution with fixed mean m and covariance C, so that $X'_t \sim N_d(m, C)$. Independence MH is one of the simplest and best understood MH Markov chains. However, it is unlikely to be effective in many settings, as will be shown later.

Another well-studied version is the *Symmetric Metropolis-Hastings* algorithm, which results when the proposal distribution is symmetric about the current state X_{t-1}, so $q(y \mid x) = q(x \mid y)$. For example, suppose the proposal is a d-dimensional normal distribution centered at the previous iteration with covariance matrix hI_d so that $X'_t \mid X_{t-1} \sim N_d(X_{t-1}, hI_d)$. The scaling parameter $h > 0$ is user-specified.

If the proposal of a symmetric Metropolis-Hastings also satisfies $q(y \mid x) = q(\|x - y\|)$, then a *Random-Walk Metropolis-Hastings* algorithm results. Note that the proposal, $X'_t \mid X_{t-1} \sim N_d(X_{t-1}, hI_d)$, in the previous example results in random walk Metropolis-Hastings. Another example would be a Uniform$(x - h, x + h)$, $h > 0$, proposal.

Suppose the proposal distribution is a d-dimensional normal distribution where the mean takes a gradient descent step and covariance hI_d so that

$$X'_t \mid X_{t-1} \sim N_d(X_{t-1} + (h/2)\nabla \log(f(X_{t-1})), hI_d).$$

The gradient does not require the normalization constant of f. The motivation for this construction comes from discretized Langevin dynamics [13] and, consequently, is known as *Metropolis-Adjusted Langevin Algorithm* (MALA).

These examples barely scratch the surface of Metropolis-Hastings variations. For example, much recent research has gone into Hamiltonian Monte Carlo which uses a proposal based on discretized Hamiltonian dynamics [11]. Extensions to Riemanian manifold MALA and Hamiltonian variants also exist [5]. Some of these will be encountered later.

4 Combining Markov Kernels

Markov kernels may be combined in order to create more effective algorithms. There are two basic ways of doing so. For now, suppose $P_1, \ldots, P_d$ are Markov kernels such that $FP_i = F$ for

$i = 1, \ldots, d$. The *composition* kernel is defined by

$$P_C(x, \cdot) = (P_1 \cdots P_d)(x, \cdot)$$

and corresponds to cycling through the kernels in a specified order. If each $r_i > 0$ such that $r_1 + \cdots + r_d = 1$, then the *mixing* kernel is defined by

$$P_m(x\cdot) = r_1 P_1(x, \cdot) + \cdots + r_d P_d(x, \cdot)$$

and corresponds to updating via the kernel selected by the mixing probabilities r_i. It is an easy exercise to verify that $FP_c = F$ and $FP_m = F$.

5 Component-wise Updates

It is rare that Metropolis-Hastings can be used without modification in practically relevant settings. The distribution F is often too complicated or too high-dimensional for a full-dimensional Metropolis-Hastings update to be effective. It is natural to seek to work with smaller problems. This can be accomplished by identifying Markov kernels associated with the marginal and conditional distributions of F and then using composition and mixing to combine them.

Since this is introductory, the focus will be on the setting where $f(x, y)$ is a density function on $\mathsf{X} \times \mathsf{Y}$, that is, the so-called *two-variable* setting. The extension to the setting with more than two variables is straightforward through the usual properties of joint probability functions, but some complications can occur as will be discussed later.

5.1 Linchpin Variables

Let $f_{X|Y}$ be the pf of the conditional distribution of X given Y. Let f_Y be the pf of the marginal distribution of Y. If sampling from $f_{X|Y}$ is straightforward, recall that Y is a linchpin variable since

$$f(x, y) = f_{X|Y}(x|y)\, f_Y(y), \tag{4}$$

and that exact samples can be obtained by first simulating $Y \sim F_Y$ followed by $X \sim F_{X|Y}$. When it is difficult to sample from F_Y directly it is natural to turn to MCMC. Let $P_Y(y, \cdot)$ be a Markov kernel such that F_Y is invariant. Then the linchpin variable sampler is specified in Algorithm 3.

Algorithm 3 Linchpin variable sampler

1: *Input:* Current value (X_j, Y_j)

2: Draw $Y_{j+1} \sim P_Y(Y_j, \cdot)$.

3: Draw $X_{j+1} \sim F_{X|Y}(\cdot \mid Y_{j+1})$.

4: Set $j = j + 1$

The resulting Markov kernel is given by

$$P_{LV}((x, y), A) = \int_A F_{X|Y}(dx' \mid y') P_Y(y, dy'). \tag{5}$$

Note that P_{LV} is a composition of two Markov kernels.

Example 5.1. Let $Y \in \mathbb{R}^n$, $\beta \in \mathbb{R}^p$, $u \in \mathbb{R}^k$, X be a known $n \times p$ full column rank design matrix, and Z a known $n \times k$ full column rank matrix. Also assume that $\max\{p, k\} < n$. Then a Bayesian linear model is given by the following hierarchy

$$\begin{aligned} Y|\beta, u, \lambda_E, \lambda_R &\sim \mathrm{N}_n\left(X\beta + Zu, \lambda_E^{-1} I_n\right) \\ g(\beta) &\propto 1 \\ u|\lambda_E, \lambda_R &\sim \mathrm{N}_k\left(0, \lambda_R^{-1} I_k\right) \\ \lambda_E &\sim \mathrm{Gamma}\left(e_1, e_2\right) \\ \lambda_R &\sim \mathrm{Gamma}\left(r_1, r_2\right) . \end{aligned} \tag{6}$$

Assume that $e_1, e_2, r_1, r_2 > 0$ are known hyper-parameters. This hierarchy results in a proper posterior [14] . Let $\xi = (\beta^T, u^T)^T$, $\lambda = (\lambda_E, \lambda_R)^T$ and let y denote all of the data. Then the posterior density satisfies

$$f(\beta, u, \lambda_E, \lambda_R \mid y) = f(\xi, \lambda \mid y) = f_{\xi|\lambda}(\xi \mid \lambda, y)\, f_\lambda(\lambda \mid y). \tag{7}$$

It is left as an exercise to determine the distributions for $\xi \mid \lambda, y$ and $\lambda \mid y$ and hence verify that λ is a linchpin variable.

Linchpin variable samplers have been employed in a variety of scenarios. Their success is typically due to either (1) superior mixing in the lower-dimensional space, (2) de-correlation of components via the linchpin variables, or (3) lower post-processing costs.

In Algorithm 3, suppose P_Y has Markov transition density $k_Y : \mathsf{Y} \times \mathsf{Y} \to [0, \infty)$ with invariant density f_Y. That is,

$$\int_{\mathsf{Y}} k_Y(y' \mid y) dy' = 1 \quad \text{and} \quad f_Y(y') = \int_{\mathsf{Y}} f_Y(y) k_Y(y' \mid y) dy.$$

Consequently, the Markov transition density for the linchpin variable sampler $(x, y) \mapsto (x', y')$ is

$$k(x', y' \mid x, y) = f_{X|Y}(x' \mid y')\, k_Y(y' \mid y). \tag{8}$$

Notice that k leaves f invariant since

$$\begin{aligned}
\int \int k(x', y' \mid x, y) f(x, y)\, dx\, dy &= \int \int f_{X|Y}(x' \mid y')\, k_Y(y' \mid y) f(x, y)\, dy\, dx \\
&= f_{X|Y}(x' \mid y') \int k_Y(y' \mid y) f_Y(y) \int f_{X|Y}(x \mid y)\, dx\, dy \\
&= f_{X|Y}(x' \mid y') f_Y(y') \int k(y \mid y')\, dy \\
&= f(x', y').
\end{aligned}$$

Also, k_Y satisfies the detailed balance condition when

$$k_Y(y' \mid y) f_Y(y) = k_Y(y \mid y') f_Y(y') \qquad \text{for all } y, y'. \tag{9}$$

Transition density k_Y satisfies detailed balance with respect to f_Y if and only if the linchpin variable sampler satisfies detailed balance with respect to f. That is, for all x', y', x, y

$$k(x', y' \mid x, y) f(x, y) = k(x, y \mid x', y') f(x', y'). \tag{10}$$

The argument is as follows. Assume (10). Let $y, y' \in \mathsf{Y}$ and let $x' \in \mathsf{X}$ be such that $f_{X|Y}(x' \mid y') > 0$.

Then

$$
\begin{aligned}
f_Y(y)k_Y(y' \mid y) &= \int_{\mathsf{X}} f(x,y)k_Y(y' \mid y)\, dx \\
&= \int_{\mathsf{X}} \frac{f(x,y)f_{X|Y}(x' \mid y')k_Y(y' \mid y)}{f_{X|Y}(x' \mid y')}\, dx \\
&= \int_{\mathsf{X}} \frac{f(x,y)k(x',y' \mid x,y)}{f_{X|Y}(x' \mid y')}\, dx \\
&= \int_{\mathsf{X}} \frac{f(x',y')k(x,y \mid x',y')}{f_{X|Y}(x' \mid y')}\, dx \\
&= \int_{\mathsf{X}} \frac{f_{X|Y}(x' \mid y')f_Y(y')f_{X|Y}(x \mid y)k_Y(y \mid y')}{f_{X|Y}(x' \mid y')}\, dx \\
&= f_Y(y')k_Y(y \mid y') \int_{\mathsf{X}} f_{X|Y}(x \mid y)\, dx \\
&= f_Y(y')k_Y(y \mid y')\,,
\end{aligned}
$$

and hence (9) holds. Now assume (9) so that,

$$
\begin{aligned}
f(x,y)k(x',y' \mid x,y) &= f(x,y)f_{X|Y}(x' \mid y')k_Y(y' \mid y) \\
&= f_{X|Y}(x \mid y)f_{X|Y}(x' \mid y')f_Y(y)k_Y(y' \mid y) \\
&= f_{X|Y}(x \mid y)f_{X|Y}(x' \mid y')f_Y(y')k_Y(y \mid y') \\
&= f(x',y')f_{X|Y}(x \mid y)k_Y(y \mid y') \\
&= f(x',y')k(x,y \mid x',y')\,,
\end{aligned}
$$

and hence (10) holds.

5.2 Gibbs Samplers

A two-variable Gibbs sampler was introduced in Example 2.4 and invariance was established in Example 2.5. More generally, let $F(\mathrm{d}x, \mathrm{d}y)$ have support $\mathsf{X} \times \mathsf{Y}$ and let $F_{X|Y}(\mathrm{d}x|y)$, $y \in \mathsf{Y}$, and $F_{Y|X}(\mathrm{d}y|x)$, $x \in \mathsf{X}$, be full conditional distributions. One version is the deterministic-scan Gibbs (DG) sampler, which is now described.

Algorithm 4 Deterministic-scan Gibbs sampler

1: *Input:* Current value $(X_n, Y_n) = (x, y)$.

2: Draw Y_{n+1} from $F_{Y|X}(\cdot \mid x)$, and call the observed value y'.

3: Draw X_{n+1} from $F_{X|Y}(\cdot \mid y')$.

4: Set $n = n + 1$.

If $f_{X|Y}(x \mid y)$ and $f_{Y|X}(y \mid x)$ are the conditional pfs associated with the joint pf $f(x, y)$, then

$$k_{DG}(x', y' \mid x, y) = f_{X|Y}(x' \mid y) f_{Y|X}(y' \mid x')$$

is the Markov transition density for the two-variable deterministic scan Gibbs sampler. Note that the Gibbs sampler is the composition of Markov kernels. The same argument as in Example 2.5 establishes the invariance of F.

The two-variable Gibbs sampler has properties not found in most other MCMC algorithms and that will be exploited later in the study of their convergence properties. In particular, the Gibbs sampler is not reversible with respect to F, but each of the marginal processes is a reversible Markov chain with respect to the appropriate marginal of F. The Y process has Markov transition density

$$k_Y(y' \mid y) = \int f_{Y|X}(y' \mid x) f_{X|Y}(x \mid y) dx,$$

which satisfies detailed balance with respect to the marginal density $f_Y(y)$

$$\begin{aligned}
k_Y(y' \mid y) f_Y(y) &= \int f_{Y|X}(y' \mid x) f_{X|Y}(x \mid y) f_Y(y) dx \\
&= \int f_{Y|X}(y' \mid x) f(x, y) dx \\
&= \int f_{Y|X}(y' \mid x) f_{Y|X}(y \mid x) f_X(x) dx \\
&= \int f_{Y|X}(y \mid x) f(x, y') dx \\
&= k_Y(y \mid y') f_Y(y')
\end{aligned}$$

A similar calculation shows that the X process satisfies detailed balance with respect to $f_X(x)$.

An alternative to the deterministic scan Gibbs sampler is the random-scan Gibbs (RG) sampler which is described below.

Algorithm 5 Random-scan Gibbs sampler with selection probability $r \in (0, 1)$

1: *Input:* Current value $(X_n, Y_n) = (x, y)$.

2: Draw $U \sim \text{Bernoulli}(r)$, and call the observed value u.

3: If $u = 1$, draw X_{n+1} from $F_{X|Y}(\cdot \mid y)$, and set $Y_{n+1} = y$.

4: If $u = 0$, draw Y_{n+1} from $F_{Y|X}(\cdot \mid x)$, and set $X_{n+1} = x$.

5: Set $n = n + 1$.

The random scan Gibbs sampler is the mixture of Markov kernels. The associated Markov transition density is

$$k_{RG}(x', y' \mid x, y) = r f_{X|Y}(x' \mid y)\delta(y - y') + (1 - r) f_{Y|X}(y' \mid x)\delta(x - x').$$

Of course, F is invariant:

$$\begin{aligned}
\iint f(x, y) k_{RG}(x', y' \mid x, y) \mathrm{d}x\mathrm{d}y =& r \iint f(x, y) f_{X|Y}(x' \mid y)\delta(y - y')\mathrm{d}x\mathrm{d}y \\
&+ (1 - r) \iint f(x, y) f_{Y|X}(y' \mid x)\delta(x - x')\mathrm{d}x\mathrm{d}y \\
=& r \int f_Y(y) f_{X|Y}(x' \mid y)\delta(y - y')\mathrm{d}y \\
&+ (1 - r) \int f_X(x) f_{Y|X}(y' \mid x)\delta(x - x')\mathrm{d}x \\
& r f_Y(y') f_{X|Y}(x' \mid y') + (1 - r) f_X(x') f_{Y|X}(y' \mid x') \\
=& f(x', y').
\end{aligned}$$

It is left as an exercise to show that the random scan Gibbs sampler is F-symmetric.

Two-component Gibbs samplers are surprisingly applicable in many Bayesian statistical models since they arise naturally in data augmentation settings [7, 15, 16]. This will be considered more carefully later.

5.3 Collapsed Gibbs Samplers

Through mixing and composition there are seemingly limitless MCMC algorithms that can be created from the those that have already been described. The collapsed Gibbs sampler is a particularly instructive example, which has proved useful in many settings [1, 2, 3, 8, 9, 12]. Here the three-component setting is considered, but it can be generalized.

Let $F(\mathrm{d}x_1, \mathrm{d}x_2, \mathrm{d}x_3)$ have support $\mathsf{X}_1 \times \mathsf{X}_2 \times \mathsf{X}_3$. Then, of course, the joint distribution can be factored as the product of a marginal and a conditional. Specifically,

$$F(\mathrm{d}x_1, \mathrm{d}x_2, \mathrm{d}x_3) = F_{X_1|X_2,X_3}(\mathrm{d}x_1 \mid x_2, x_3) F_{X_2,X_3}(\mathrm{d}x_2, \mathrm{d}x_3),$$

which suggests the following collapsed Gibbs sampler algorithm. That is, the algorithm combines a two-variable Gibbs sampler for the marginal $F_{X_2,X_3}(\mathrm{d}x_2, \mathrm{d}x_3)$, with draws from the conditional.

Algorithm 6 Collapsed (Deterministic-scan) Gibbs sampler

1: *Input:* Current value $(X_{1,n}, X_{2,n}, X_{3,n})$.

2: Draw $X_{2,n+1} \sim F_{X_2|X_3}(\cdot \mid x_{3,n})$.

3: Draw $X_{3,n+1} \sim F_{X_3|X_2}(\cdot \mid x_{2,n+1})$.

4: Draw $X_{1,n+1} \sim F_{X_1|X_2,X_3}(\cdot \mid x_{2,n+1}, x_{3,n+1})$

5: Set $n = n + 1$.

If the transition density for the Gibbs sampler with invariant density $f_{X_2,X_3}(x_2, x_3)$ is

$$k_{DG}(x_2', x_3' \mid x_2, x_3) = f_{X_2|X_3}(x_2' \mid x_3) f_{X_3|X_2}(x_3' \mid x_2'),$$

then the Markov transition density for the collapsed Gibbs sampler is

$$k_{CG}(x_1', x_2', x_3' \mid x_1, x_2, x_3) = f_{x_1|x_2,x_3}(x_1' \mid x_2', x_3') k_{DG}(x_2', x_3' \mid x_2, x_3).$$

Notice that f is invariant for the collapsed Gibbs sampler since

$$\begin{aligned}
&\iiint f(x_1,x_2,x_3)k_{CG}(x_1',x_2',x_3' \mid x_1,x_2,x_3)\mathrm{d}x_1\mathrm{d}x_2\mathrm{d}x_3\\
=&\iiint f(x_1,x_2,x_3)f_{x_1|x_2,x_3}(x_1' \mid x_2',x_3')k_{DG}(x_2',x_3' \mid x_2,x_3)\mathrm{d}x_1\mathrm{d}x_2\mathrm{d}x_3\\
=&f_{x_1|x_2,x_3}(x_1' \mid x_2',x_3')f_{X_3|X_2}(x_3' \mid x_2')\iint f_{X_2,X_3}(x_2,x_3)f_{X_2|X_3}(x_2' \mid x_3)\mathrm{d}x_2\mathrm{d}x_3\\
=&f_{x_1|x_2,x_3}(x_1' \mid x_2',x_3')f_{X_3|X_2}(x_3' \mid x_2')f_{X_2}(x_2')\\
=&f(x_1',x_2',x_3').
\end{aligned}$$

Example 5.2. Consider a Bayesian vector autogression model. For $p > 0$, let $X_1, X_2, \ldots$ be a p-vector of predictors, and let $\epsilon_1, \epsilon_2, \ldots$ be independent and identically distributed according to a $N(0, \Sigma)$, where Σ is an $r \times r$ positive-definite matrix. Let $B \in \mathbb{R}^{r\times r}$ and for some $q \geq 1$ and $i = 1, \ldots, q$ let $A_i \in \mathbb{R}^{r\times r}$ such that,

$$Y_t = \sum_{i=1}^{q} A_i^T Y_{t-1} + B^T X_t + \epsilon_t\,.$$

The process $\{X_t\}$ is independent of $\{\epsilon_t\}$. Let $A = [A_1^T, A_2^T, \ldots A_q^T]^T \in \mathbb{R}^{qr\times r}$, and let all the data observed until time K be $D = \{(X_1, Y_1), (X_2, Y_2), \ldots, (X_K, Y_K)\}$. For hyper-parameters $m \in \mathbb{R}^{qr^2}$, C, a $qr^2 \times qr^2$ positive-definite matrix, and D, an $r\times r$ positive-definite matrix, the three parameters are given the following independent priors:

$$\begin{aligned}
f(\mathrm{vec}(A)) &\propto \exp\left\{-\frac{1}{2}[\mathrm{vec}(A) - m]^T C[\mathrm{vec}(A) - m]\right\}\,,\\
f(\Sigma) &\propto \det(\Sigma)^{-a/2} \exp\left\{\mathrm{tr}\left(-\frac{1}{2}D\Sigma^{-1}\right)\right\} \quad \text{and}\,,\\
f(B) &\propto 1\,.
\end{aligned}$$

There are well-known conditions for which $(A, B, \Sigma) \mid D$ is a proper distribution [3]. It is left as an exercise to show that the conditionals $A \mid \Sigma, D$ and $\Sigma \mid A, D$ are available to sample and hence a collapsed Gibbs sampler can be constructed with transition density

$$k_{CG}(A', B', \Sigma' \mid A, B, \Sigma) = f(B' \mid A', \Sigma', D)f(A' \mid \Sigma', D)\, f(\Sigma' \mid A, D).$$

5.4 Conditional Samplers

A common type of MCMC algorithm in practically relevant applications is the conditional Metropolis-Hastings (CMH) sampler. These arise when it is infeasible to sample from at least one of the conditional distributions associated with F so that at least one Metropolis-Hastings update must be used. Assume that $F_{Y|X}$ and $F_{X|Y}$, respectively, admit density functions $f_{Y|X}$ and $f_{X|Y}$. Let $q(\cdot \mid x, y)$, $(x, y) \in \mathsf{X} \times \mathsf{Y}$, be a proposal function on X. A deterministic-scan CMH (DC) sampler is now described, but a more general algorithm will be encountered later.

Algorithm 7 Deterministic-scan CMH sampler

1: *Input:* Current value $(X_n, Y_n) = (x, y)$

2: Draw Y_{n+1} from $F_{Y|X}(\cdot \mid x)$, and call the observed value y'.

3: Draw a random element Z from $q(\cdot \mid x, y')$, and call the observed value z. With probability

$$a(z; x, y') = \min\left\{1, \frac{f_{X|Y}(z \mid y')q(x \mid z, y')}{f_{X|Y}(x \mid y')q(z \mid x, y')}\right\},$$

set $X_{n+1} = z$; with probability $1 - a(z; x, y')$, set $X_{n+1} = x$.

4: Set $n = n + 1$.

Set

$$r(x, y') = 1 - \int_{\mathsf{X}} q(z \mid x, y')\alpha(z, x, y')\, \mu_X(dz)\,.$$

Then define

$$k_{MH}(x' \mid x, y') = q(x' \mid x, y')\alpha(x', x, y') + \delta(x' - x)r(x, y')$$

so that, by construction, the corresponding kernel is reversible with respect to $f_{X|Y}(\cdot|y')$ for each y'. If

$$k_{CMH}(x', y' \mid x, y) = f_{Y|X}(y' \mid x)k_{MH}(x' \mid x, y'),$$

then the transition kernel for CMH is

$$P_{CMH}((x, y), A) = \int_A k_{CMH}(x', y' \mid x, y)\, \mu(d(x', y')) \qquad A \in \mathcal{B}(\mathsf{X}) \times \mathcal{B}(\mathsf{Y})\,.$$

It is left as an exercise to show that $FP_{CMH} = F$, but P_{CMH} is not reversible with respect to F.

There is an obvious alternative random-scan CMH (RC) sampler.

Algorithm 8 Random-scan CMH sampler with selection probability $r \in (0,1)$

1: *Input:* Current value $(X_n, Y_n) = (x, y)$.

2: Draw $U \sim \text{Bernoulli}(r)$, and call the observed value u.

3: If $u = 1$, draw a random element Z from $q(\cdot \mid x, y)$, and call the observed value z. With probability

$$a(z; x, y) = \min\left\{1, \frac{f_{X|Y}(z \mid y) q(x \mid z, y)}{f_{X|Y}(x \mid y) q(z \mid x, y)}\right\},$$

set $X_{n+1} = z$; with probability $1 - a(z; x, y)$, set $X_{n+1} = x$. Set $Y_{n+1} = y$.

4: If $u = 0$, draw Y_{n+1} from $F_{Y|X}(\cdot \mid x)$, and set $X_{n+1} = x$.

5: Set $n = n + 1$.

Fix $p \in (0,1)$ and let

$$k_{RCMH}(x', y' \mid x, y) = p f_{Y|X}(y'|x)\delta(x' - x) + (1-p) k_{MH}(x' \mid x, y)\delta(y' - y) ,$$

then the transition kernel for is

$$P_{RCMH}((x,y), A) = \int_A k_{RCMH}(x', y' \mid x, y)\, \mu(d(x', y')) \qquad A \in \mathcal{B}(\mathsf{X}) \times \mathcal{B}(\mathsf{Y}) .$$

It is left as an exercise to show that P_{RCMH} is reversible with respect to F.

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